Complex Analysis Homework 1

张溢致

2026 年 3 月 9 日

1. Prove that

$$\int_0^\infty \sin(x^2)\,dx = \int_0^\infty \cos(x^2)\,dx = \frac{\sqrt{2\pi}}{4}.$$

These are the Fresnel integrals. Here, $\int_0^\infty$ is interpreted as $\lim_{R\to\infty}\int_0^R$.

[Hint: Integrate the function e^{-z^2} over the path in Figure 14. Recall that $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.]

证明 考虑复函数 $f(z) = e^{-z^2}$，积分路径取第一象限中由以下三段组成的闭合曲线 Γ_R：

1. 实轴上的线段 C_1：$z = x$，$x \in [0, R]$；
2. 圆弧 C_2：$z = Re^{i\theta}$，$\theta \in [0, \pi/4]$；
3. 射线 C_3：$z = re^{i\pi/4}$，$r \in [0, R]$。

由柯西积分定理：

$$\int_{C_1} e^{-z^2}dz + \int_{C_2} e^{-z^2}dz - \int_{C_3} e^{-z^2}dz = 0.$$

对于第一段：

$$\int_{C_1} e^{-z^2}dz = \int_0^R e^{-x^2}dx \xrightarrow{R\to\infty} \frac{\sqrt{\pi}}{2}.$$

对于第二段：令 $z = Re^{i\theta}$，则 $z^2 = R^2e^{i2\theta}$，$dz = iRe^{i\theta}d\theta$，

$$\left|\int_{C_2} e^{-z^2}dz\right| \le \int_0^{\pi/4} |e^{-R^2(\cos 2\theta + i\sin 2\theta)}| \cdot Rd\theta = R\int_0^{\pi/4} e^{-R^2\cos 2\theta}d\theta.$$

当 $\theta \in [0, \pi/4]$ 时，$\cos 2\theta \ge 1 - 4\theta/\pi$，因此：

$$R\int_0^{\pi/4} e^{-R^2(1-4\theta/\pi)}d\theta = Re^{-R^2}\int_0^{\pi/4} e^{4R^2\theta/\pi}d\theta = \frac{\pi}{4R}e^{-R^2}(e^{R^2}-1) \xrightarrow{R\to\infty} 0.$$

对于第三段：令 $z = re^{i\pi/4}$，则 $z^2 = ir^2$，$dz = e^{i\pi/4}dr$，

$$\int_{C_3} e^{-z^2}dz = e^{i\pi/4}\int_0^R e^{-ir^2}dr = \frac{\sqrt{2}}{2}(1+i)\int_0^R (\cos r^2 - i\sin r^2)dr.$$

令 $R \to \infty$，得：

$$\frac{\sqrt{\pi}}{2} = \frac{\sqrt{2}}{2}(1+i)\left(\int_0^\infty \cos x^2 dx - i\int_0^\infty \sin x^2 dx\right).$$

$$(1-i)\frac{\sqrt{2\pi}}{4} = \int_0^\infty \cos x^2 dx - i\int_0^\infty \sin x^2 dx.$$

对比实部虚部即得证。

2. Show that $\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$.

[Hint: The integral equals $\frac{1}{2i}\int_{-\infty}^{\infty} \frac{e^{ix}-1}{x}\,dx$. Use the indented semicircle.]

证明 考虑复函数 $f(z) = \frac{e^{iz}}{z}$，积分路径取上半平面的挖去原点的半圆 Γ_R^ϵ $(R \to \infty, \epsilon \to 0)$：

1. 实轴上的线段 C_1：$x \in [\epsilon, R]$；
2. 上半圆弧 C_2：$z = Re^{i\theta}$，$\theta \in [0, \pi]$；
3. 实轴上的线段 C_3：$x \in [-R, -\epsilon]$；
4. 小半圆 C_4：$z = \epsilon e^{i\theta}$，$\theta \in [\pi, 0]$。

由柯西积分定理：

$$\int_{C_1} \frac{e^{iz}}{z}dz + \int_{C_2} \frac{e^{iz}}{z}dz + \int_{C_3} \frac{e^{iz}}{z}dz - \int_{C_4} \frac{e^{iz}}{z}dz = 0.$$

当 $R \to \infty$ 时：

$$\int_{C_2} \frac{e^{iz}}{z}dz \xrightarrow{R\to\infty} 0.$$

令 $z = \epsilon e^{i\theta}$，则 $dz = i\epsilon e^{i\theta}d\theta$，

$$\int_{C_4} \frac{e^{iz}}{z}dz = \int_\pi^0 \frac{e^{i\epsilon e^{i\theta}}}{\epsilon e^{i\theta}} \cdot i\epsilon e^{i\theta}d\theta = i\int_\pi^0 (1+o(1))d\theta \xrightarrow{\epsilon\to 0} i\pi.$$

而

$$\int_{C_1} \frac{e^{iz}}{z}dz + \int_{C_3} \frac{e^{iz}}{z}dz = \int_\epsilon^R \frac{e^{ix}}{x}dx + \int_{-R}^{-\epsilon} \frac{e^{ix}}{x}dx = \int_\epsilon^R \frac{e^{ix}-e^{-ix}}{x}dx = 2i\int_\epsilon^R \frac{\sin x}{x}dx.$$

令 $R \to \infty, \epsilon \to 0$，得：

$$2i\int_0^\infty \frac{\sin x}{x}dx - i\pi = 0 \implies \int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}.$$

证毕。

7. Suppose $f : \mathbb{D} \to \mathbb{C}$ is holomorphic. Show that the diameter $d = \sup_{z,w\in\mathbb{D}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \le d.$$

Moreover, it can be shown that equality holds precisely when f is linear, $f(z) = a_0 + a_1 z$.

Note. In connection with this result, see the relationship between the diameter of a curve and Fourier series described in Problem 1, Chapter 4, Book I.

[Hint: $2f'(0) = \frac{1}{2\pi i}\int_{|\zeta|=r} \frac{f(\zeta)-f(-\zeta)}{\zeta^2}\, d\zeta$ whenever $0 < r < 1$.]

证明 对于任意 $0 < r < 1$，使得:

$$2f'(0) = \frac{1}{2\pi i}\int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2}\, d\zeta.$$

令 $\zeta = re^{i\theta}$，则 $d\zeta = ire^{i\theta}d\theta$，代入得:

$$2f'(0) = \frac{1}{2\pi}\int_0^{2\pi} \frac{f(re^{i\theta}) - f(-re^{i\theta})}{r^2 e^{i2\theta}} \cdot ire^{i\theta} d\theta = \frac{1}{2\pi r}\int_0^{2\pi} [f(re^{i\theta}) - f(-re^{i\theta})]e^{-i\theta} d\theta.$$

对上述等式取模:

$$2|f'(0)| \le \frac{1}{2\pi r}\int_0^{2\pi} |f(re^{i\theta}) - f(-re^{i\theta})| d\theta \le \frac{d}{r}$$

由于对于任意 $0 < r < 1$ 上述不等式都成立，于是令 $r \to 1$，有

$$2|f'(0)| \le d$$

若等号成立，则对所有 θ, $|f(e^{i\theta}) - f(-e^{i\theta})| = d$。设 $f(z) = \sum_{n=0}^{\infty} a_n z^n$，则 $f(\zeta) - f(-\zeta) = 2\sum_{k=0}^{\infty} a_{2k+1}\zeta^{2k+1}$，代入 $\int_{|\zeta|=r} \frac{f(\zeta)-f(-\zeta)}{\zeta^2}\, d\zeta$:

$$2f'(0) = 2a_1 = \frac{1}{2\pi i}\int_{|\zeta|=r} \frac{2\sum_{k=0}^{\infty} a_{2k+1}\zeta^{2k+1}}{\zeta^2} d\zeta$$

若 f 非线性，则存在 $n \ge 2$ 使得 $a_n \ne 0$，此时 $|f(\zeta) - f(-\zeta)| < d$，等号不成立。故等号当且仅当 $f(z) = a_0 + a_1 z$ 时成立。证毕。

8. If f is a holomorphic function on the strip $-1 < y < 1$, $x \in \mathbb{R}$ with

$$|f(z)| \le A(1 + |z|)^{\eta}, \quad \eta \text{ a fixed real number}$$

for all z in that strip, show that for each integer $n \geq 0$ there exists $A_n \geq 0$ so that

$$|f^{(n)}(x)| \leq A_n(1+|x|)^\eta, \quad \text{for all } x \in \mathbb{R}.$$

[Hint: Use the Cauchy inequalities.]

证明 由柯西不等式，对任意 $x \in \mathbb{R}$，取以 x 为中心、半径为 $1/2$ 的圆周 $C(x, 1/2)$，则：

$$|f^{(n)}(x)| \leq \frac{n!}{(1/2)^n} \cdot \sup_{z \in C(x,1/2)} |f(z)|.$$

对 $z \in C(x, 1/2)$，有 $|z| \leq |x| + 1/2 \leq 2(1+|x|)$，因此：

$$|f(z)| \leq A(1+|z|)^\eta \leq A(1+2(1+|x|))^\eta \leq A(3(1+|x|))^\eta = A \cdot 3^\eta (1+|x|)^\eta.$$

令 $A_n = n! \cdot 2^n \cdot A \cdot 3^\eta$，则：

$$|f^{(n)}(x)| \leq n! \cdot 2^n \cdot A \cdot 3^\eta (1+|x|)^\eta = A_n(1+|x|)^\eta.$$

9. Let Ω be a bounded open subset of $\mathbb{C}$, and $\varphi : \Omega \to \Omega$ a holomorphic function. Prove that if there exists a point $z_0 \in \Omega$ such that

$$\varphi(z_0) = z_0 \quad \text{and} \quad \varphi'(z_0) = 1$$

then φ is linear.

[Hint: Why can one assume that $z_0 = 0$? Write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ near 0, and prove that if $\varphi_k = \varphi \circ \cdots \circ \varphi$ (where φ appears k times), then $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$. Apply the Cauchy inequalities and let $k \to \infty$ to conclude the proof. Here we use the standard O notation, where $f(z) = O(g(z))$ as $z \to 0$ means that $|f(z)| \leq C|g(z)|$ for some constant C as $|z| \to 0$.]

证明 令 $\psi(z) = \varphi(z + z_0) - z_0$，则 $\psi(0) = 0$，$\psi'(0) = \varphi'(z_0) = 1$，且 ψ 映有界开集 $\Omega - z_0$ 到自身。只需证明 $\psi(z) = z$，则 $\varphi(z) = z - z_0 + z_0 = z$。

设 $\psi(z) = z + a_n z^n + O(z^{n+1})$ $(n \geq 2,\ a_n \neq 0)$，记 $\psi_k = \psi \circ \psi \circ \cdots \circ \psi$，用数学归纳法证明：

$$\psi_k(z) = z + k a_n z^n + O(z^{n+1}).$$

基例：$k = 1$，显然成立；

归纳：若 $\psi_{k-1}(z)=z+(k-1)a_nz^n+O(z^{n+1})$，则：

$$\psi_k(z)=\psi(\psi_{k-1}(z))=\psi_{k-1}(z)+a_n(\psi_{k-1}(z))^n+O((\psi_{k-1}(z))^{n+1})=z+ka_nz^n+O(z^{n+1}).$$

因 $\Omega-z_0$ 有界，存在 $M>0$ 使得 $|\psi_k(z)|\le M$ 对所有 $z\in\Omega-z_0$ 成立。取以 0 为中心、半径 r 的圆周 $C_r\subset\Omega-z_0$，由柯西不等式：

$$|ka_n|=|\frac{\psi_k^{(n)}(0)}{n!}|\le\frac{n!M}{r^n}.$$

即 $|a_n|\le\frac{M}{kr^n}$，令 $k\to\infty$，得 $a_n=0$，与假设矛盾。

因此 $\psi(z)=z$，即 $\varphi(z)=z$。证毕。

11. Let f be a holomorphic function on the disc D_{R_0} centered at the origin and of radius R_0.

(a) Prove that whenever $0<R<R_0$ and $|z|<R$, then

$$f(z)=\frac{1}{2\pi}\int_0^{2\pi}f(Re^{i\varphi})\,\mathrm{Re}\left(\frac{Re^{i\varphi}+z}{Re^{i\varphi}-z}\right)d\varphi.$$

(b) Show that

$$\mathrm{Re}\left(\frac{Re^{i\gamma}+r}{Re^{i\gamma}-r}\right)=\frac{R^2-r^2}{R^2-2Rr\cos\gamma+r^2}.$$

[Hint: For the first part, note that if $w=R^2/\bar{z}$, then the integral of $f(\zeta)/(\zeta-w)$ around the circle of radius R centered at the origin is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired identity.]

证明 *(a)* 由柯西积分公式，$f(z)=\frac{1}{2\pi i}\int_{|\zeta|=R}\frac{f(\zeta)}{\zeta-z}d\zeta$。令 $w=R^2/\bar{z}$，则 $|w|>R$，故 $\int_{|\zeta|=R}\frac{f(\zeta)}{\zeta-w}d\zeta=0$。

两式相减得：

$$f(z)=\frac{1}{2\pi i}\int_{|\zeta|=R}f(\zeta)\left(\frac{1}{\zeta-z}-\frac{1}{\zeta-w}\right)d\zeta$$

计算得：

$$\begin{aligned}\frac{1}{\zeta-z}-\frac{1}{\zeta-w}&=\frac{1}{\zeta-z}-\frac{\overline{z}}{\overline{z}\zeta-\zeta\overline{\zeta}}\\&=\frac{\zeta\overline{\zeta}-z\overline{z}}{\zeta(\zeta-z)(\overline{\zeta}-\overline{z})}\end{aligned}$$

而

$$\begin{aligned}\mathrm{Re}\left(\frac{\zeta+z}{\zeta-z}\right) &= \frac{\zeta+z}{\zeta-z}+\frac{\overline{\zeta+z}}{\overline{\zeta-z}}\\ &= \frac{\zeta\overline{\zeta}-z\overline{z}}{(\zeta-z)(\overline{\zeta}-\overline{z})}\end{aligned}$$

于是

$$f(z)=\frac{1}{2\pi i}\int_{|\zeta|=R} f(\zeta)\left(\frac{1}{\zeta-z}-\frac{1}{\zeta-w}\right)d\zeta=\frac{1}{2\pi i}\int_{|\zeta|=R}\frac{1}{\zeta}f(\zeta)\,\mathrm{Re}\left(\frac{\zeta+z}{\zeta-z}\right)d\zeta$$

令 $\zeta=Re^{i\varphi}$，则 $d\zeta=iRe^{i\varphi}d\varphi$，代入得：

$$f(z)=\frac{1}{2\pi}\int_0^{2\pi} f(Re^{i\varphi})\,\mathrm{Re}\left(\frac{Re^{i\varphi}+z}{Re^{i\varphi}-w}\right)d\varphi$$

(b) 上一问中得到

$$\mathrm{Re}\left(\frac{\zeta+z}{\zeta-z}\right)=\frac{\zeta\overline{\zeta}-z\overline{z}}{(\zeta-z)(\overline{\zeta}-\overline{z})}$$

代入 $\zeta=Re^{i\gamma}, z=r$ 得

$$\mathrm{Re}\left(\frac{Re^{i\gamma}+r}{Re^{i\gamma}-r}\right)=\frac{R^2-r^2}{R^2+r^2-rR(e^{i\gamma}+e^{-i\gamma})}=\frac{R^2-r^2}{R^2-2Rr\cos\gamma+r^2}$$

12. Let u be a real-valued function defined on the unit disc $\mathbb{D}$. Suppose that u is twice continuously differentiable and harmonic, that is,

$$\Delta u(x,y)=0$$

for all $(x,y)\in\mathbb{D}$.

(a) Prove that there exists a holomorphic function f on the unit disc such that

$$\mathrm{Re}(f)=u.$$

Also show that the imaginary part of f is uniquely defined up to an additive (real) constant.

[Hint: From the previous chapter we would have $f'(z)=2\partial u/\partial z$. Therefore, let $g(z)=2\partial u/\partial z$ and prove that g is holomorphic. Why can one find F with $F'=g$? Prove that $\mathrm{Re}(F)$ differs from u by a real constant.]

(b) Deduce from this result, and from Exercise 11, the Poisson integral representation formula from the Cauchy integral formula: If u is harmonic in the unit disc and continuous on its closure, then if $z = re^{i\theta}$ one has

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi)\, d\varphi$$

where $P_r(\gamma)$ is the Poisson kernel for the unit disc given by

$$P_r(\gamma) = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2}.$$

证明 *(a)* 令 $g(z) = 2\frac{\partial u}{\partial z} = u_x - iu_y$，由调和性 $\Delta u = u_{xx} + u_{yy} = 0$ 和 u 二阶微分的存在，故 g 满足柯西-黎曼方程，故 g 全纯。

因 $\mathbb{D}$ 单连通，存在全纯函数 F 使得 $F' = g$，则 $\mathrm{Re}(F') = u_x$，$\mathrm{Im}(F') = -u_y$。令 $v = \mathrm{Im}(F)$，则 $v_x = -\mathrm{Re}(F') = -u_x$，$v_y = \mathrm{Im}(F') = -u_y$，故 $u - \mathrm{Re}(F)$ 的偏导数均为 *0*，即 $u = \mathrm{Re}(F) + C$（C 为实常数）。

令 $f(z) = F(z) + C$，则 $\mathrm{Re}(f) = u$。若另有全纯函数 f_1 使得 $\mathrm{Re}(f_1) = u$，则 $f - f_1$ 为纯虚函数，故为虚常数，从而虚部唯一（最多相差一个实常数）。

(b) 由 *(a)*，存在全纯函数 f 使得 $\mathrm{Re}(f) = u$。对 f 应用第 *11* 题的结果：

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\varphi})\, \mathrm{Re}\left(\frac{e^{i\varphi} + z}{e^{i\varphi} - z}\right) d\varphi.$$

取实部得：

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{i\varphi}) \cdot \frac{1 - r^2}{1 - 2r\cos(\theta - \varphi) + r^2} d\varphi = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi) d\varphi.$$

证毕。

13. Suppose f is an analytic function defined everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in the expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is equal to 0. Prove that f is a polynomial.

[Hint: Use the fact that $c_n n! = f^{(n)}(z_0)$ and use a countability argument.]

证明 由泰勒展开系数公式，$c_n = \frac{f^{(n)}(z_0)}{n!}$，于是条件转化为对于任意 z_0，存在 n_0 使得 $f^{(n_0)}(z_0) = 0$

记 $E_n = \{z \in \mathbb{C} \mid f^{(n)}(z) = 0\}$，则 $\bigcup_{n=0}^{\infty} E_n = \mathbb{C}$。因 $f^{(n)}$ 全纯，故 E_n 为闭集。由贝尔纲定理，$\mathbb{C}$ 作为完备度量空间，必有某个 E_N 包含非空开集 U。

在 U 上，$f^{(N)}(z) = 0$，故 $f^{(N)}$ 在 $\mathbb{C}$ 上恒为 0，因此 f 为次数不超过 $N-1$ 的多项式。证毕。

14. Suppose that f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

denotes the power series expansion of f in the open unit disc, then

$$\lim_{n\to\infty} \frac{a_n}{a_{n+1}} = z_0.$$

证明 由第三章定理 *1.2*: 若 z_0 是 f 的极点，则存在唯一正整数 m 和 z_0 邻域内非零全纯函数 $h(z)$，使得

$$f(z) = (z - z_0)^{-m} h(z).$$

因 f 在包含闭单位圆盘的开集上除 z_0 外全纯，故 $h(z)$ 可延拓为该开集上的全纯函数，且 $h(z_0) \neq 0$。

对 $|z| < 1$，展开有：

$$\frac{1}{z - z_0} = -\frac{1}{z_0} \cdot \frac{1}{1 - \dfrac{z}{z_0}} = -\frac{1}{z_0} \sum_{k=0}^{\infty} \left(\frac{z}{z_0}\right)^k = -\sum_{k=0}^{\infty} \frac{z^k}{z_0^{k+1}}.$$

对其求 $m-1$ 阶导数：

$$\frac{1}{(z - z_0)^m} = \frac{(-1)^m}{(m-1)!} \cdot \frac{d^{m-1}}{dz^{m-1}} \left(\frac{1}{z - z_0}\right) = \sum_{k=0}^{\infty} \binom{k+m-1}{m-1} \cdot \frac{(-1)^m}{z_0^{k+m}} z^k.$$

记 z^k 的系数为 d_k，即

$$d_k = \binom{k+m-1}{m-1} \frac{(-1)^m}{z_0^{k+m}}.$$

因 $h(z)$ 在包含闭单位圆盘的开集上全纯，其幂级数展开为：

$$h(z)=\sum_{n=0}^{\infty}b_nz^n,$$

且收敛半径 $R>1$，故存在常数 M 使得 $|b_n|\le MR^{-n}$。

于是 $f(z)=\left(\sum_{k=0}^{\infty}d_kz^k\right)\left(\sum_{n=0}^{\infty}b_nz^n\right)=\sum_{n=0}^{\infty}a_nz^n$，其中

$$a_n=\sum_{k=0}^{n}d_kb_{n-k}=\sum_{j=0}^{n}d_{n-j}b_j.$$

代入 d_k 的表达式得

$$a_n=\frac{(-1)^m}{z_0^m}\sum_{j=0}^{n}\binom{n-j+m-1}{m-1}z_0^{-(n-j)}b_j=\frac{(-1)^m}{z_0^{n+m}}\sum_{j=0}^{n}\binom{n-j+m-1}{m-1}z_0^jb_j.$$

令

$$S_n=\sum_{j=0}^{n}\binom{n-j+m-1}{m-1}z_0^jb_j.$$

由于 $|b_j|\le MR^{-j}$ 且 $\binom{n-j+m-1}{m-1}\le C(n-j)^{m-1}\le Cn^{m-1}$，对任意固定的 N 有

$$\left|\sum_{j=N+1}^{n}\binom{n-j+m-1}{m-1}z_0^jb_j\right|\le Cn^{m-1}\sum_{j=N+1}^{\infty}R^{-j}=Cn^{m-1}\frac{R^{-N}}{1-1/R}.$$

取 N 充分大可使此项相对于 n^{m-1} 任意小。而对 $j\le N$，当 $n\to\infty$ 时，

$$\binom{n-j+m-1}{m-1}=\binom{n+m-1}{m-1}\left(1+O\left(\frac{1}{n}\right)\right),$$

因此

$$\sum_{j=0}^{N}\binom{n-j+m-1}{m-1}z_0^jb_j=\binom{n+m-1}{m-1}\sum_{j=0}^{N}z_0^jb_j+O(n^{m-2}).$$

令 $N\to\infty$，得

$$S_n=\binom{n+m-1}{m-1}h(z_0)+o(n^{m-1}),$$

其中 $h(z_0)=\sum_{j=0}^{\infty}b_jz_0^j\neq0$。于是

$$a_n=\frac{(-1)^m}{z_0^{n+m}}\left(\binom{n+m-1}{m-1}h(z_0)+o(n^{m-1})\right).$$

同理，

$$a_{n+1} = \frac{(-1)^m}{z_0^{n+1+m}} \left(\binom{n+m}{m-1} h(z_0) + o(n^{m-1}) \right).$$

因此

$$\frac{a_n}{a_{n+1}} = z_0 \cdot \frac{\binom{n+m-1}{m-1}}{\binom{n+m}{m-1}} \cdot \frac{1+o(1)}{1+o(1)}.$$

而

$$\frac{\binom{n+m-1}{m-1}}{\binom{n+m}{m-1}} = \frac{(n+m-1)!}{(m-1)!\,n!} \cdot \frac{(m-1)!\,(n+1)!}{(n+m)!} = \frac{n+1}{n+m} \to 1,$$

故 $\lim\limits_{n\to\infty} \dfrac{a_n}{a_{n+1}} = z_0$。

1. Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C. A point w on C is said to be regular for f if there is an open neighborhood U of w and an analytic function g on U, so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f.

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \quad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc.

[Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \to \infty$ as $r \to 1$.]

(b) Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \quad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle.

[Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

证明 *(a)* 取 $\theta = 2\pi p/2^k$ $(p, k$ 为正整数$)$，令 $z = re^{i\theta}$，则 $z^{2^n} = r^{2^n} e^{i2^n\theta} = r^{2^n} e^{i2\pi p 2^{n-k}}$。

当 $n \geq k$ 时，$e^{i2\pi p 2^{n-k}} = 1$，故：

$$|f(re^{i\theta})| = \left| \sum_{n=0}^{k-1} r^{2^n} e^{i2^n\theta} + \sum_{n=k}^{\infty} r^{2^n} \right| \geq \sum_{n=k}^{\infty} r^{2^n} - \sum_{n=0}^{k-1} r^{2^n}.$$

令 $r \to 1^-$，右侧趋向 ∞，故 $f(z)$ 在这些点附近无界，无法解析延拓到单位圆周外。

(b) 首先，级数 $\sum_{n=0}^{\infty} 2^{-n\alpha}|z|^{2^n} \leq \sum_{n=0}^{\infty} 2^{-n\alpha} = \frac{1}{1-2^{-\alpha}}$，收敛，故 $f(z)$ 连续到单位圆周。

类似 *(a)*，取 $\theta = 2\pi p/2^k$，则：

$$|f(re^{i\theta})| \geq \sum_{n=k}^{\infty} 2^{-n\alpha} r^{2^n} - \sum_{n=0}^{k-1} 2^{-n\alpha} r^{2^n}.$$

令 $r \to 1^-$，右侧趋向 $\sum_{n=k}^{\infty} 2^{-n\alpha} = \frac{2^{-k\alpha}}{1-2^{-\alpha}} > 0$，且 $f(z)$ 在这些点的导数无界，故无法解析延拓。证毕。